

■ 3.6.6 Solving Equations Involving Power Series

`LogicalExpand[series1 == series2]`

give the equations obtained by equating corresponding coefficients in the power series

`Solve[series1 == series2, {a1, a2, ...}]`

solve for coefficients in power series

Solving equations involving power series.

Here is a power series.

`In[1]:= y = 1 + Sum[a[i] x^i, {i, 3}] + 0[x]^4`

`Out[1]= 1 + a[1] x + a[2] x^2 + a[3] x^3 + 0[x]^4`

This gives an equation involving the power series.

`In[2]:= D[y, x]^2 - y == x`

`Out[2]= (-1 + a[1]^2) + (-a[1] + 4 a[1] a[2]) x +
(-a[2] + 4 a[2]^2 + 6 a[1] a[3]) x^2 + 0[x]^3 == x`

`LogicalExpand` generates a sequence of equations for each power of x .

`In[3]:= LogicalExpand[ % ]`

`Out[3]= -1 + a[1]^2 == 0 &&
-1 - a[1] + 4 a[1] a[2] == 0 &&
-a[2] + 4 a[2]^2 + 6 a[1] a[3] == 0`

This solves the equations for the coefficients $a[i]$. You can also feed equations involving power series directly to `Solve`.

`In[4]:= Solve[ % ]`

`Out[4]= {{a[3] -> -(1/12), a[2] -> 1/2, a[1] -> 1},
{a[3] -> 0, a[2] -> 0, a[1] -> -1}}`

Some equations involving power series can also be solved using the `InverseSeries` function discussed on page 513.